

NFL Natural-Language Search — Test Suite

Use this document to validate that the Text-to-SQL feature works correctly.

For each test: enter the question in the Smart Search page, then compare the result to the Expected Answer column. Mark Pass / Fail / Partial in the Status column.

How to run these tests

1. Open the deployed app (or local dev) → Smart Search page
2. Enter the question exactly as written
3. Note the SQL generated (enable "Show SQL" toggle — see Phase 2.5 in the work plan)
4. Compare result to Expected Answer
5. If the SQL is wrong or result is wrong, note what went wrong in the Notes column

Pass criteria:

- The SQL is valid PostgreSQL and executes without error
- The result matches the expected answer (within ± 1 for rounding)
- No extra/missing rows

Category 1 — Basic career totals (should use *_career views)

#	Question	Expected answer	Table used	Status	Notes
1	How many career passing yards does Tom Brady have?	89,214 yards	passing_career		
2	How many career touchdown passes did Peyton Manning throw?	539 TDs	passing_career		
3	What is Jerry Rice's career receiving touchdowns total?	197 TDs	offense_career		
4	How many career rushing yards does Emmitt Smith have?	18,355 yards	offense_career		
5	What are Adrian Peterson's career rushing yards?	14,820 yards	offense_career		

#	Question	Expected answer	Table used	Status	Notes
6	How many career sacks does Bruce Smith have?	200 sacks	<code>defense_career</code>		
7	What is Adam Vinatieri's career field goals made total?	599 FGM	<code>kicking_career</code>		

Category 2 — Single-season queries (should use `*_seasons` + `season =` filter)

#	Question	Expected answer	Table used	Status	Notes
8	How many touchdowns did Patrick Mahomes throw in 2022?	41 TDs	<code>passing_seasons</code> WHERE season=2022		
9	Who led the NFL in rushing yards in 2012?	Adrian Peterson, 2,097 yards	<code>offense_seasons</code> WHERE season=2012		
10	How many receiving yards did Justin Jefferson have in 2022?	1,809 yards	<code>offense_seasons</code> WHERE season=2022		
11	Who had the most sacks in 2023?	Micah Parsons / check actual DB value	<code>defense_seasons</code> WHERE season=2023		
12	How many passing yards did Joe Burrow throw for in 2021?	4,611 yards	<code>passing_seasons</code> WHERE season=2021		
13	What was Tyreek Hill's receiving yards total in 2023?	Check actual DB value	<code>offense_seasons</code> WHERE season=2023		

Category 3 — Leaderboards (should use ORDER BY + LIMIT)

#	Question	Expected answer	Notes
14	Who are the top 5 QBs by career passing yards?	Brady, Brees, Favre, Manning, Roethlisberger (approx order)	Should hit <code>passing_career</code>

#	Question	Expected answer	Notes
15	Who are the top 10 receivers by career receiving touchdowns?	Rice at top, Moss, Gronk near top	<code>offense_career</code>
16	Which kicker has the highest career field goals made?	Vinatieri or Andersen depending on coverage	<code>kicking_career</code>
17	Who has the most career sacks in the database?	Bruce Smith or Reggie White	<code>defense_career</code>
18	Top 5 rushers by career rushing yards	Smith, Faulk, Peterson, Barry, Payton (approx)	<code>offense_career</code>
19	Which QB threw the most TDs in a single season in the last 10 years?	Peyton Manning 2013 (55) or check	<code>passing_seasons</code> WHERE season >= 2015

Category 4 — Position filters

#	Question	Expected answer	Notes
20	Who is the best wide receiver in the 2023 season by receiving yards?	Check DB	Should filter <code>pos = 'WR'</code> in <code>offense_seasons</code>
21	List all tight ends drafted in the first round since 2015	Should include e.g. T.J. Hockenson, Kyle Pitts	<code>draft</code> WHERE pos='TE' AND round=1 AND draft_year >= 2015
22	Which running backs have the most rushing TDs in 2022?	Derrick Henry near top	<code>offense_seasons</code> WHERE season=2022, filter by position from players
23	Who are the top defensive players by interceptions since 2018?	Check DB	<code>defense_seasons</code> WHERE season >= 2018, ORDER BY int

Category 5 — Cross-table JOINS (harder — tests whether Claude joins correctly)

#	Question	Expected answer	Notes
24	What pick was Patrick Mahomes in the draft?	10th overall, 2017, Round 1	<code>draft</code> JOIN <code>players</code> on <code>player_id</code>
25	How tall is Lamar Jackson?	6'2" (stored as "6_2")	<code>combine_seasons</code> JOIN <code>players</code>
26	How many career TDs did the 2017 first-round picks score collectively?	Check DB	<code>draft</code> WHERE <code>draft_year=2017</code> AND <code>round=1</code> JOIN <code>offense_career</code>
27	Which players had more than 1000 receiving yards in 2022 and were drafted after 2018?	Check DB	<code>offense_seasons</code> JOIN <code>draft</code> via <code>players</code>
28	What was Josh Allen's snap percentage in 2022?	Check DB	<code>snap_counts</code> WHERE <code>season=2022</code> , JOIN <code>players</code>
29	Which QBs have the best CPOE since 2020?	Check <code>ngs_passing</code> , Mahomes/Allen area	<code>ngs_passing</code> WHERE <code>season >= 2020</code>

Category 6 — Tricky / edge cases (likely to fail — important to catch)

#	Question	Notes on what to watch for
30	What is Tom Brady's career completion percentage?	⚠️ Must NOT use a <code>cmp_pct</code> column from <code>*_career</code> (it doesn't exist). Must compute <code>100.0 * SUM(cmp) / SUM(att)</code> . Check the SQL.
31	Who is the tallest QB ever drafted?	⚠️ Must use <code>split_part(ht, '_', 1)::int * 12 + split_part(ht, '_', 2)::int</code> to convert height. If it returns garbage, Claude is not handling the underscore format.
32	How many games did Will Campbell play as a rookie?	⚠️ Campbell appears in <code>draft</code> but has null <code>player_id</code> (no box-score stats yet). Should return 0 or "no tracked stats" — NOT an error.
33	What is Patrick Mahomes' yards per attempt in his career?	⚠️ Must compute from <code>SUM(yds)/SUM(att)</code> — not use a non-existent career column.

#	Question	Notes on what to watch for
34	Who won the most Super Bowls?	⚠️ Super Bowl data is not in this database. Should return CANNOT_ANSWER or a clear "this data isn't available" message.
35	Who is the best player of all time?	⚠️ Unanswerable, subjective. Should return CANNOT_ANSWER.
36	How many players are in the database?	Should count <code>players</code> table: ~11,627. Easy sanity check.
37	Which team drafted the most players in the 2020 draft?	Should GROUP BY team in <code>draft</code> WHERE draft_year=2020, ORDER BY count DESC.
38	What's Antonio Brown's drop rate in 2018?	Should use <code>adv_receiving</code> table (coverage starts 2018). Tests whether Claude uses supplementary tables.
39	Which RB had the best rush yards over expected in 2021?	Should use <code>ngs_rushing.ryoe_per_att</code> WHERE season=2021. Tests NGS table awareness.
40	How many times was Aaron Donald listed as questionable in 2019?	Should use <code>injuries</code> WHERE player_id = Donald's id AND season=2019 AND report_status='Questionable'.

Category 7 — Hebrew language queries (בדיקת שפה עברית)

#	Question	Expected
41	כמה יארדים זרק טום ברייד בקריירה?	89,214 — same as test #1
42	מי היה הרצ'ן הטוב ביותר של 2022 לפי יארדים?	Justin Jefferson
43	כמה סאקים עשה באד לי בקריירה?	Look up in DB
44	מי הם 5 הרצ'נים הטובים ביותר בקריירה?	Same as test #15
45	מה גובהו של להמר ג'קסון?	6'2"

Scoring guide

After running all tests, calculate:

Pass rate by category:

Category 1 (basic career): ___ / 7

Category 2 (single season): ___ / 6

Category 3 (leaderboards): ___ / 6

Category 4 (position filter): ___ / 4

Category 5 (cross-table): ___ / 6

Category 6 (edge cases): ___ / 11

Category 7 (Hebrew): ___ / 5

Total: ___ / 45

What to do with failures:

- If categories 1–3 fail a lot → the system prompt schema description is wrong or incomplete
- If category 5 fails → Claude isn't joining tables correctly; add JOIN examples to the prompt
- If category 6 (edge cases) fail → specific rules need to be added to the system prompt
- If category 7 fails → check that the prompt explicitly says Hebrew is supported

Regression test — run after every prompt change

These 8 questions must always pass. If any of them break after a prompt edit, revert.

#	Question
---	----------

R1	How many career passing yards does Tom Brady have?
----	--

R2	Who led the NFL in rushing yards in 2012?
----	---

R3	What is Tom Brady's career completion percentage?
----	---

R4	Who are the top 5 QBs by career passing yards?
----	--

R5	What pick was Patrick Mahomes in the draft?
----	---

R6	Who won the Super Bowl? (should return CANNOT_ANSWER)
----	---

R7	How many players are in the database?
----	---------------------------------------

R8	כמה יארדים זרק טום ברייד בקריירה?
----	-----------------------------------